The general case, $C^{\mathrm{ps}}(g,d)$

Contributing maps here are enumerated by pairs (g_1,g_2) with $g_1+g_2=g$ and $g_1,g_2\geqslant 0$. These are ordered partitions of g. We study the general case g_1,g_2 with $g_1\neq g-1,0$ and we treat these ones separately.

The (g_1,g_2) **contribution**

The curve $C_{(g_1,g_2)}$ has the normalization sequence

$$0\to\mathcal{O}_C\to\mathcal{O}_{g_1}\oplus\mathcal{O}_{\mathbb{P}^1}\oplus\mathcal{O}_{g_2}\to\underline{\mathbb{C}}_{n_1}\oplus\underline{\mathbb{C}}_{n_2}\to 0.$$

Tensoring with $\bullet := f^*\mathcal{O}(-H+s+t)$ gives

$$0\to\bullet\to\mathcal{O}(s)\oplus\mathcal{O}(-d\widetilde{H}+s+t)\oplus\mathcal{O}(s+t)\to\langle(s)\rangle\oplus\langle(s+t)\rangle\to 0.$$

After taking the long exact sequence in cohomology and simplifying, we're left with

$$0\to H^0(\mathcal{O}(s))\oplus H^0(\mathcal{O}(s+t))\to H^0(\langle(s)\rangle)\oplus H^0(\langle(s+t)\rangle)$$
$$\to H^1(\bullet)\to H^1(\mathcal{O}(s))\oplus H^1(\mathcal{O}(-d\widetilde{H}+s+t))\oplus H^1(\mathcal{O}(s+t))\to 0$$

Taking Euler classes gives us

$$\begin{aligned}e(H^1(\bullet))&=e\big(H^1(\mathcal{O}(s))\big)e\big(H^1(\mathcal{O}(-d\widetilde{H}+s+t))\big)e\big(H^1(\mathcal{O}(s+t))\big)\\&=(-1)^{g_1}\Lambda^{\mathrm{ps}}_{g_1}(-s)\left(\prod_{k=1}^{d-1}\left(s+\frac{kt}{d}\right)\right)(-1)^{g_2}\Lambda^{\mathrm{ps}}_{g_2}(-(s+t))\end{aligned}$$

We find the numerator by plugging in $s=-t$, $s=0$ and multiplying.

$$\begin{aligned}&i^*_{(g_1,g_2)}(e(H^1f^*\mathcal{O}(-H))e(H^1f^*\mathcal{O}(-H+t)))\\&=(-1)^{g_1}\Lambda^{\mathrm{ps}}_{g_1}(t)\prod_{k=1}^{d-1}\left(-t+\frac{kt}{d}\right)(-1)^{g_2}\lambda^{\mathrm{ps}}_{g_2}(-1)^{g_1}\lambda^{\mathrm{ps}}_{g_1}\prod_{k=1}^{d-1}\left(\frac{kt}{d}\right)(-1)^{g_2}\Lambda^{\mathrm{ps}}_{g_2}(-t)\\&=\big((-1)^{d-1}(t/d)^{2d-2}(d-1)!^2\big)\big(\lambda^{\mathrm{ps}}_{g_1}\Lambda^{\mathrm{ps}}_{g_1}(t)\big)\big(\lambda^{\mathrm{ps}}_{g_2}\Lambda^{\mathrm{ps}}_{g_2}(-t)\big)\end{aligned}$$

There's no automorphisms as there's no smooth points so that the normal bundle only considers deformations and obstructions. In this case

$$e(\mathrm{Def}(C))=(-\psi_{P_1}-\psi_{Q_1})(-\psi_{P_2}-\psi_{Q_2})=\left(-\psi_{P_1}+\frac{t}{d}\right)\left(-\psi_{P_2}+\frac{-t}{d}\right)$$

For the map, we tensor the normalization sequence by $f^*T\mathbb{P}^1$ to get

$$0\to f^*T\mathbb{P}^1\to\mathcal{O}(t)\oplus\mathcal{O}(2d\widetilde{H}-t)\oplus\mathcal{O}(-t)\to\langle(t)\rangle\oplus\langle(-t)\rangle\to 0.$$

Taking cohomology gets us

$$0 \to \mathrm{Def}(f) \to H^0(\mathcal{O}(t)) \oplus H^0(\mathcal{O}(2d\widetilde{H}-t)) \oplus H^0(\mathcal{O}(-t)) \to H^0(\langle (t) \rangle) \oplus$$

$$\hookrightarrow \mathrm{Ob}(f) \longrightarrow H^1(\mathcal{O}(t)) \oplus H^1(\mathcal{O}(-t)) \longrightarrow 0$$

thus the desired quotient of Euler classes is

$$\frac{e(\mathrm{Def}(f))}{e(\mathrm{Ob}(f))} = \frac{e(H^0(\mathcal{O}(2d\widetilde{H}-t)))}{e(H^1(\mathcal{O}(t)))e(H^1(\mathcal{O}(-t)))} = \frac{(t/d)^{2d} d!^2 (-1)^d}{(-1)^{g_1}\Lambda^{\mathrm{ps}}_{g_1}(-t)(-1)^{g_2}\Lambda^{\mathrm{ps}}_{g_2}(t)}.$$

We end by condensing everything into the integrand.

$$\begin{aligned}
&\frac{\left((-1)^{d-1}(t/d)^{2d-2}(d-1)!^2\right)\left(\lambda^{\mathrm{ps}}_{g_1}\Lambda^{\mathrm{ps}}_{g_1}(t)\right)\left(\lambda^{\mathrm{ps}}_{g_2}\Lambda^{\mathrm{ps}}_{g_2}(-t)\right)(-1)^{g_1}\Lambda^{\mathrm{ps}}_{g_1}(-t)(-1)}{(t/d)^{2d} d!^2 (-1)^d(-\psi_{P_1}+t/d)(-\psi_{P_2}+(-t/d))} \\
=&\frac{\left(\lambda^{\mathrm{ps}}_{g_1}\Lambda^{\mathrm{ps}}_{g_1}(t)(-1)^{g_1}\Lambda^{\mathrm{ps}}_{g_1}(-t)\right)\left(\lambda^{\mathrm{ps}}_{g_2}\Lambda^{\mathrm{ps}}_{g_2}(-t)(-1)^{g_2}\Lambda^{\mathrm{ps}}_{g_2}(t)\right)}{(-t^2)(-\psi_{P_1}+t/d)(-\psi_{P_2}+(-t/d))} \\
=&d^2\left(\frac{\lambda^{\mathrm{ps}}_{g_1}\Lambda^{\mathrm{ps}}_{g_1}(t)(-1)^{g_1}\Lambda^{\mathrm{ps}}_{g_1}(-t)}{t^2(1-d\psi_{P_1}/t)}\right)\left(\frac{\lambda^{\mathrm{ps}}_{g_2}\Lambda^{\mathrm{ps}}_{g_2}(-t)(-1)^{g_2}\Lambda^{\mathrm{ps}}_{g_2}(t)}{t^2(1-(-d\psi_{P_2}/t))}\right).
\end{aligned}$$

We've seen before the minus sign on the psi class has no effect because of even powers. In this case, we just treat a general term of genus g. Expanding via Matt's relation (theorem **??**) gets us to

$$\begin{aligned}
&\frac{\lambda^{\mathrm{ps}}_{g}\Lambda^{\mathrm{ps}}_{g}(t)(-1)^{g}\Lambda^{\mathrm{ps}}_{g}(-t)}{t^2(1-d\psi_P/t)} \\
=&\frac{\lambda^{\mathrm{ps}}_{g}}{t^2(1-d\psi_P/t)}\sum_{k=0}^{g}\frac{t^{2g-2k}}{k!}\mathrm{gl}^k_*\left(\prod_{j=1}^{k}\psi_{\star j}-\psi_{\bullet j}\right) \\
=&\sum_{k=0}^{g}\frac{d^{2g-2k-2}}{k!}\lambda^{\mathrm{ps}}_{g}\mathrm{gl}^k_*\left(\prod_{j=1}^{k}\psi_{\star j}-\psi_{\bullet j}\right)\psi_P^{2g-2k-2} \\
=&\sum_{k=0}^{g}\frac{d^{2g-2k-2}}{k!}\mathrm{gl}^k_*\left(p_0^*(\lambda_{g-k})\prod_{j=1}^{k}\psi_{\star j}p_j^*(\lambda_1)\right)\psi_P^{2g-2k-2}
\end{aligned}$$

Exercise 0.0.1. The product

$$\lambda^{\mathrm{ps}}_{g}\mathrm{gl}^k_*\left(\prod_{j=1}^{k}\psi_{\star j}-\psi_{\bullet j}\right)\psi_P^{2g-2k-2}$$

results in

$$\lambda_g\mathrm{gl}^k_*\left(\prod_{j=1}^{k}\psi_{\star j}-\psi_{\bullet j}\right)\psi_P^{2g-2k-2} = \mathrm{gl}^k_*\left(p_0^*(\lambda_{g-k})\prod_{j=1}^{k}\psi_{\star j}p_j^*(\lambda_1)\right)\psi_P^{2g-2k-2}.$$

However on its own,

$$\lambda_g^{\mathrm{ps}}\mathrm{gl}_*^k\left(\prod_{j=1}^k \psi_{\star j}-\psi_{\bullet j}\right) \neq \mathrm{gl}_*^k\left(p_0^*(\lambda_{g-k})\prod_{j=1}^k \psi_{\star j}p_j^*(\lambda_1)\right).$$

Observe that the difference is that we're not multiplying the class $\psi_P^{2g-2k-2}$. To that effect do the following:

(a) Describe a general formula for the product

$$\lambda_g^{\mathrm{ps}}\Lambda_g^{\mathrm{ps}}(t)(-1)^g\Lambda_g^{\mathrm{ps}}(-t).$$

(b) Verify that indeed after multiplying the ψ class $\psi_P^{2g-2k-2}$, we get

$$\lambda_g^{\mathrm{ps}}\mathrm{gl}_*^k\left(\prod_{j=1}^k \psi_{\star j}-\psi_{\bullet j}\right)\psi_P^{2g-2k-2} = \lambda_g\mathrm{gl}_*^k\left(\prod_{j=1}^k \psi_{\star j}-\psi_{\bullet j}\right)\psi_P^{2g-2k-2},$$

which means that all terms in the expansion of λ_g^{ps} besides the first one go to zero when multiplied by

$$\mathrm{gl}_*^k\left(\prod_{j=1}^k \psi_{\star j}-\psi_{\bullet j}\right)\psi_P^{2g-2k-2}.$$

We may normalize $t=1$ use the expansion of λ_g^{ps} **??** and the expansion of the total lambda classes **??**, and then distribute the product so that we get a sum of terms of the form

$$\mathrm{gl}_*^i(p_0(\lambda_{g-i}))\cdot\mathrm{gl}_*^j((\psi_{\star 1}-\psi_{\bullet 1})\cdots(\psi_{\star j}-\psi_{\bullet j}))$$

Let us color the edges of these graphs blue and green respectively in order to color the result.

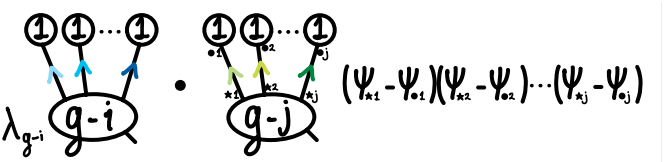

Figure 0.1: Initial coloring of product

Call $m=\min(i,j)$ and $M=\max(i,j)$, the product will result in a sum of graphs with the structure of $\mathrm{gl}_*^{M+k}(1)$. In order to color its edges we first color with the one associated with the smaller index, m. There are

$$\binom{M+k}{m}m! \text{ ways to color with } m \text{ colors.}$$

$(M+k)-m$ edges remain unpainted. We have to pick

$$M-((M+k)-m)=m-k$$

edges to paint with the color corresponding to M. Since the colors are distinguishable, we pick

$$\binom{M}{m-k}$$

for the overlap. And there's

$$\binom{m}{m-k}(m-k)!$$

ways to bicolor. Finally the remaining edges can be painted in $(M-m+k)!$ ways. Let Bic be the set of bicolored edges, This is wrong as I'm assuming there's more green edges right nowand G the set of green edges. Our resulting sum of graphs is:

$$\sum_{k=M}^{\min(g,i+j)} a_k \mathrm{gl}_*^{M+k}\left(p_0^*(\lambda_{g-(M+k)})\prod_{j\in \mathrm{Bic}}(-\psi_{\star j}^2)\prod_{j\in \mathrm{G}}(\psi_{\star j}p_j^*\lambda_1)\right)$$

where a_k is the coefficient counting the number of ways to color the graph divided by the automorphisms of the graph.

$$a_k = \frac{\binom{M+k}{m}m!\binom{M}{m-k}\binom{m}{m-k}(m-k)!(M-m+k)!}{(M+k)!}$$

"

Exercise 0.0.2.

Taking the integral over $\overline{M}_{g,1}$ gets us to

$$\sum_{k=0}^{g}\frac{d^{2g-2k-2}}{k!}\left(\int_{\overline{M}_{g-k,1+k}}\lambda_{g-k}\prod_{j=1}^{k}\psi_{\star j}\psi_P^{2g-2k-2}\right)\left(\int_{\overline{M}_{1,1}}\lambda_1\right)^k$$
$$=\sum_{k=0}^{g}\frac{d^{2g-2k-2}}{k!}\frac{(2g-2-k)!}{(2g-2-2k)!}b_{g-k}\frac{1}{24^k}$$

Here we applied the λ_g-conjecture **??** again to find the value of the first integral. Call this amount I_g. Then our desired (g_1,g_2) contribution is

$$I_{(g_1,g_2)} = d^2 I_{g_1} I_{g_2}.$$

The $(g,0)$ contribution

The only difference with the previous calculation lies in the automorphisms bundle. In this case we get an

$$e(\mathrm{Aut}(C)) = -t/d$$

which cancels with the t/d from the curve deformations, and the minus sign cancels out the minus in the $-t^2$. On the $(0,g)$ case, signs

are inverted but the only change is in the psi class in the denominator which doesn't alter the result of the integral. Thus

$$I_{(g,0)} = I_{(0,g)} = I_g$$

Exercise 0.0.3. Write this neatly.

The $(g-1,1)$ contribution

Exercise 0.0.4. Prove that

$$I_{(g-1,1)} = I_{(1,g-1)} = \frac{d^2-1}{24} I_{g-1}.$$

The whole contribution and generating functions

The full quantity we are looking for is

$$\begin{aligned} C^{\mathrm{ps}}(g,d) &= \frac{1}{d}\left(I_{(g,0)} + I_{(g-1,1)} + \cdots + I_{(g_1,g_2)} + \cdots + I_{(1,g-1)} + I_{(0,g)}\right) \\ &= \frac{1}{d}\left(I_g + \frac{d^2-1}{24} I_{g-1} + \cdots + d^2 I_{g_1} I_{g_2} + \cdots + \frac{d^2-1}{24} I_{g-1} + I_g\right) \end{aligned}$$

Question. If such integrals can be extended beyond their just obtained definition, what are I_0 and I_1? Can we fit them into the sum?

Observe that

$$\begin{aligned} I_0 &= \sum_{k=0}^{0} \frac{d^{2(0)-2k-2}}{k!} \frac{(2(0)-2-k)!}{(2(0)-2-2k)!} b_{0-k} \frac{1}{24^k} \\ &= \frac{d^{-2(0)-2}}{0!} \frac{(-2-0)!}{(-2-2(0))!} b_0 \frac{1}{24^0} \\ &= \frac{b_0}{d^2} = \frac{1}{d^2} \end{aligned}$$

where we have $b_0 = 1$ (without using the Bernoulli number definition) and naïvely cancelled the $(-2)!$ on top with the one on the bottom.
On the other hand, for I_1 we compute the two terms of the sum as

$$
\begin{aligned}
I_1 &= \sum_{k=0}^{1} \frac{d^{-2k}}{k!} \frac{(-k)!}{(-2k)!} b_{1-k} \frac{1}{24^k} \\
&= \frac{d^{-2(0)}}{0!} \frac{0!}{0!} b_1 \frac{1}{24^0} + \frac{d^{-2(1)}}{1!} \frac{(-1)!}{(-2)!} b_0 \frac{1}{24} \\
&= b_1 + \frac{1}{d^2}(-1)\frac{1}{24} \\
&= \frac{1}{24} - \frac{1}{24d^2} = \frac{d^2-1}{24d^2}
\end{aligned}
$$

and here we have abused the factorials in the sense that

$$\lim_{n\to -1} \frac{n!}{(n-1)!} = \lim_{n\to -1} n = -1.$$

In this case, $I_1 = \frac{d^2-1}{24d^2}$.

Given these two conditions

$$\begin{cases} I_0 = 1/d^2, \\ I_1 = (d^2-1)/(24d^2) \end{cases} \quad \text{and} \quad \begin{cases} I_{(g,0)} = I_g, \\ I_{(g-1,1)} = \dfrac{d^2-1}{24} I_{g-1}, \end{cases}$$

these match perfectly when extending the relation $I_{(g_1,g_2)} = d^2 I_{g_1} I_{g_2}$. Observe that

$$
\begin{aligned}
I_{(g,0)} &= d^2 I_g I_0 = d^2 I_g \left(\frac{1}{d^2}\right) = I_g, \\
I_{(g-1,1)} &= d^2 I_{g-1} I_1 = d^2 I_{g-1} \left(\frac{d^2-1}{24d^2}\right) = \frac{d^2-1}{24} I_{g-1}.
\end{aligned}
$$

Theorem 0.0.5. *The generating function for the integrals I_g is*

$$\frac{1}{d^2}\left(1 - \frac{t^2}{24}\right) B\left(\frac{dt}{1-t^2/24}\right),$$

where $B(t) = \frac{t/2}{\sin\left(t/2\right)}$ is the generating function for b_g.

Proof

Let us call $\mathcal{I}(t) = \sum_{g\geqslant 0} I_g t^{2g}$, then rewriting I_g we have

$$\mathcal{I}(t) = \sum_{g\geqslant 0} \sum_{k=0}^{g} \frac{d^{2g-2k-2}}{k!} \frac{(2g-2-k)!}{(2g-2-2k)!} b_{g-k} \frac{1}{24^k} t^{2g}.$$

We apply Fubini's theorem to rearrange the sums as

$$\sum_{k\geqslant 0} \sum_{g=k}^{\infty} \frac{d^{2g-2k-2}}{k!} \frac{(2g-2-k)!}{(2g-2-2k)!} b_{g-k} \frac{1}{24^k} t^{2g}$$

and then take the change of variables $g=n+k$. From this, as $g\to k$, $n\to 0$ and as g grows, n also grows. Our sum becomes

$$\sum_{k\geqslant 0}\sum_{n=0}^{\infty}\frac{d^{2n-2}}{k!}\frac{(2n-2+k)!}{(2n-2)!}b_n\frac{1}{24^k}t^{2n+2k}.$$

We may switch order of summation again and manipulate the sum as follows

$$\begin{aligned}&\sum_{n=0}^{\infty}\sum_{k=0}^{\infty}\frac{d^{2n-2}}{k!}\frac{(2n-2+k)!}{(2n-2)!}b_n\frac{1}{24^k}t^{2n+2k}\\ =&\sum_{n=0}^{\infty}d^{2n-2}b_nt^{2n}\sum_{k=0}^{\infty}\frac{(2n-2+k)!}{(2n-2)!k!}\left(\frac{t^2}{24}\right)^k\\ =&\sum_{n=0}^{\infty}d^{2n-2}b_nt^{2n}\sum_{k=0}^{\infty}\binom{2n-2+k}{k}\left(\frac{t^2}{24}\right)^k.\end{aligned}$$

The inner sum is very suggestive of the generating function

$$\frac{1}{(1-x)^{m+1}}\sum_{k=0}^{\infty}\binom{m+k}{k}x^k,$$

so that we may rewrite it as

$$\begin{aligned}&\sum_{n=0}^{\infty}d^{2n-2}b_nt^{2n}\frac{1}{\left(1-t^2/24\right)^{(2n-2)+1}}\\ =&\frac{1}{d^2}\left(1-\frac{t^2}{24}\right)\sum_{n=0}^{\infty}b_n\left(\frac{dt}{1-t^2/24}\right)^{2n}\\ =&\frac{1}{d^2}\left(1-\frac{t^2}{24}\right)B\left(\frac{dt}{1-t^2/24}\right).\end{aligned}$$

This provees the desired identity.

Lemma 0.0.6. *The mth derivative of $1/(1-x)$ is $\frac{m!}{(1-x)^{m+1}}$ and its generating function is*

$$\sum_{k=0}^{\infty}\binom{m+k}{k}m!x^k.$$

Proof

The first part is proved by induction, then

$$\begin{aligned}
D^m \frac{1}{1-x} &= \sum_{n=m}^{\infty} n(n-1)\cdots(n-m+1)x^{n-m} \\
&= \sum_{k=0}^{\infty} (k+m)(k+m-1)\cdots(k+1)x^k \\
\Rightarrow \frac{1}{(1-x)^{m+1}} &= \sum_{k=0}^{\infty} \frac{(k+m)(k+m-1)\cdots(k+1)}{m!} x^k \\
&= \sum_{k=0}^{\infty} \frac{(k+m)!}{k!m!} x^k \\
&= \sum_{k=0}^{\infty} \binom{m+k}{k} x^k
\end{aligned}$$

We may further simplify the generating function by using the expression for $B(t)$ as

$$\mathcal{I}(t) = \frac{1}{d^2}\left(1-\frac{t^2}{24}\right)\frac{\frac{dt/2}{1-t^2/24}}{\sin\left(\frac{dt}{2(1-t^2/24)}\right)} = \frac{t/2d}{\sin\left(\frac{dt/2}{1-t^2/24}\right)},$$

whose first few terms of the expansion are

$$\begin{aligned}
&\frac{1}{d^2} + \frac{(d^2-1)t^2}{24d^2} + \frac{(7d^2+10)t^4}{5760} + \frac{(31d^4+147d^2+70)t^6}{967680} \\
&+ \frac{(381d^6+3100d^4+5880d^2+1400)t^8}{464486400} \\
&+ \frac{(2555d^8+29337d^6+102300d^4+107800d^2+15400)t^{10}}{122624409600} + \ldots
\end{aligned}$$

In order to obtain a generating function for $C^{\text{ps}}(g,d)$, we recall that

$$\begin{aligned}
C^{\text{ps}}(g,d) &= \frac{1}{d}\left(I_{(g,0)} + I_{(g-1,1)} + \cdots + I_{(g_1,g_2)} + \cdots + I_{(1,g-1)} + I_{(0,g)}\right) \\
&= \frac{1}{d}\left(d^2 I_g I_0 + d^2 I_{g-1} I_1 + \cdots + d^2 I_{g_1} I_{g_2} + \cdots + d^2 I_1 I_{g-1} + d^2 I_0 I_g\right) \\
&= d(I_g I_0 + I_{g-1} I_1 + \cdots + I_{g_1} I_{g_2} + \cdots + I_1 I_{g-1} + I_0 I_g)
\end{aligned}$$

so taking the square of $\mathcal{I}(t)$ and multiplying by d gets us

$$
\begin{aligned}
d\mathcal{I}(t)^2 &= d\sum_{g\geqslant 0}\left(\sum_{\substack{g_1+g_2=g\\ g_1,g_2\geqslant 0}} I_{g_1}I_{g_2}\right)t^{2g}\\
&= \sum_{g\geqslant 0} d\left(\sum_{\substack{g_1+g_2=g\\ g_1,g_2\geqslant 0}} I_{g_1}I_{g_2}\right)t^{2g}\\
\Rightarrow d\mathcal{I}(t)^2 &= \sum_{g\geqslant 0} C^{\mathrm{ps}}(g,d)t^{2g}\\
\Rightarrow \mathcal{C}(t) &= d\left(\frac{t/2d}{\sin\left(\frac{dt/2}{1-t^2/24}\right)}\right)^2 = \frac{t^2/4d}{\sin^2\left(\frac{dt/2}{1-t^2/24}\right)}
\end{aligned}
$$

The first few terms of this sum are

$$
\frac{1}{d^3}+\frac{(d^2-1)t^2}{12d^3}+\left(\frac{1}{576d^3}+\frac{d}{240}\right)t^4+\frac{d(10d^2+21)t^6}{60480}
$$
$$
+\frac{d(84d^4+400d^2+315)t^8}{14515200}+\frac{d(180d^6+1386d^4+2750d^2+1155)t^{10}}{958003200}+\ldots
$$